# Physics XI

**Time Allowed : 30 Minutes ______________________________ Maximum Marks : 30**

Please read the instructions carefully. You will be alloted 5 minutes specifically for this purpose.

**Instructions**

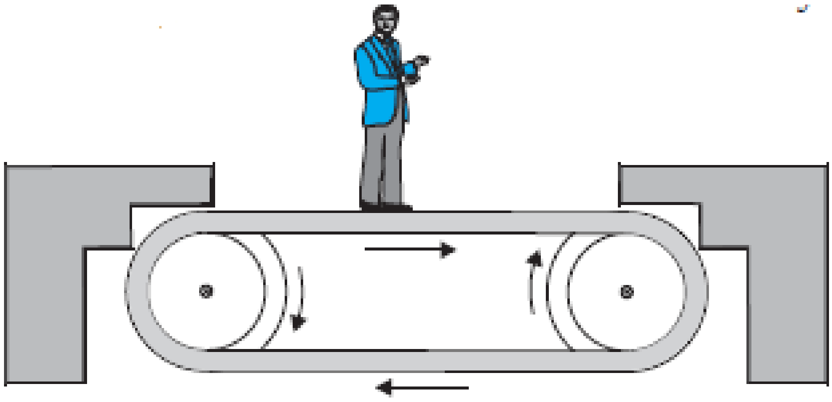
## A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

## B. Question paper format and Marking Scheme :

1. This question paper consists of 3 questions carrying 10 marks each.

**Q1:** Figure shows a man standing stationary with respect to a horizontal conveyor belt that is accelerating with $1ms^{-2}$. What is the net force on the man? If the coefficient of static friction between the man's shoes and the belt is 0.2, up to what acceleration of the belt can the man continue to be stationary relative to the belt ? (Mass of the man = 65 kg.)



**Q2:** A thin circular loop of radius R rotates about its vertical diameter with an angular frequency ω. Show that a small bead on the wire loop remains at its lowermost point for $\omega \leq \sqrt{g/R}$ . What is the angle made by the radius vector joining the centre to the bead with the vertical downward direction for $\omega = \sqrt{2g/R}$ ? Neglect friction.

**Q3:** A monkey of mass 40 kg climbs on a rope which can stand a maximum tension of 600 N. In which of the following cases will the rope break: the monkey

**(a)** climbs up with an acceleration of 6 m $s^{-2}$

**(b)** climbs down with an acceleration of 4 m $s^{-2}$

**(c)** climbs up with a uniform speed of 5 m $s^{-1}$

**(d)** falls down the rope nearly freely under gravity? (Ignore the mass of the rope).